高级工程数学

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本次内容

复习/拓展

Ch. 5

优化（开始部分）

复习

Ch. 3. Transformations

3.1 Linear Transformations

3.2 Eigenvalues and Eigenvectors

3.3 Orthogonal Projections

3.4 Quadratic Forms

3.5 Matrix Norms

Exercises

上次课例题

习题 3.5 对 $A = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 5 & 2 & 1 \\ -1 & 1 & 0 & 3 \end{bmatrix}$ 给出的 linear transform, 找一组基, 使在这组基下 A 对应的矩阵是对角阵.

解: 该题意思是:$\boldsymbol{A}$ 代表一个变换, 将 $\boldsymbol{x}$ 变成 $\boldsymbol{y}$, 即 $A\boldsymbol{x} = \boldsymbol{y}$(这里 A 就是变换对应的矩阵,$\boldsymbol{A}$,$\boldsymbol{x}$,$\boldsymbol{y}$, 均在自然基下表示).

该题问, 如果换了一组基, 在新基下 $\boldsymbol{A}$ 对应的矩阵如记成 $\boldsymbol{B}$, 要使 $\boldsymbol{B}$ 是对角阵, 基应如何取法.

claim: 新基就是 $\boldsymbol{A}$ 的特征向量 $\left\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}, ...\boldsymbol{v_n}\right\}$(如果他们构成基),$\boldsymbol{B}$ 就是 $\boldsymbol{A}$ 的特征值构成的 $\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$.

理由: $\boldsymbol{x}$ 在 $\{\boldsymbol{e}_1,\cdots \boldsymbol{e}_n\}$ 下的坐标是 $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, 在 $\{\boldsymbol{v}_1,\cdots \boldsymbol{v}_n\}$ 下坐标 $\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$.

$\boldsymbol{y}$ 在 $\{\boldsymbol{e}_1,\cdots \boldsymbol{e}_n\}$ 下的坐标是 $\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$, 在 $\{\boldsymbol{v}_1,\cdots \boldsymbol{v}_n\}$ 下坐标 $\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$.

则 $\{\boldsymbol{e_1},\cdots \boldsymbol{e_n}\}\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \{\boldsymbol{v}_1,\cdots \boldsymbol{v}_n\}\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \Longrightarrow \boldsymbol{V}\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$.(1)

$\{\boldsymbol{e}_1,\cdots \boldsymbol{e}_n\}\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \boldsymbol{V}\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$ (2)

对 (1) 两端左乘 $\boldsymbol{A}$, 得

$$\boldsymbol{A}\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \boldsymbol{AV}\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \boldsymbol{V}\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

而 $\boldsymbol{A}\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \boldsymbol{y} = \boldsymbol{V}\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$(由（2）式可得).

所以，$\boldsymbol{V}\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \boldsymbol{V}\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$

$$\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

所以在 $\boldsymbol{V}$ 下,$\boldsymbol{A}$ 对应的 $\boldsymbol{B}$ 是对角阵.

上次补充例题

$\vec{x}$，经正交变换（$Q^TQ=I$），变成$\vec{y}$，

问$\vec{y}$的长度和$\vec{x}$的长度的关系。

解：看$\vec{y}=Q\vec{x}$的长度可通过内积。

$$\langle Q\vec{x}, Q\vec{x}\rangle = (Q\vec{x})^T Q\vec{x} = \vec{x}^T Q^T Q \vec{x}$$

3.4 Quadratic Forms

A quadratic form $f : \mathbb{R}^n \to \mathbb{R}$ is a function

$$f(\boldsymbol{x}) = \boldsymbol{x}^\top Q \boldsymbol{x},$$

where Q is an $n \times n$ real matrix.

假设 Q 是对称的.

称 quadratic form $\boldsymbol{x}^\top Q \boldsymbol{x}$ **是** positive definite, if $\boldsymbol{x}^\top Q \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x}$. Semidefinite if $\boldsymbol{x}^\top Q \boldsymbol{x} \geq 0$ for all nonzero $\boldsymbol{x}$.

Negative definite, negative semidefinite.

3.5 Matrix Norms

Using arguments similar to the above, we can deduce the following important inequalities.

Rayleigh's Inequalities. If an $n \times n$ matrix $\boldsymbol{P}$ is real symmetric positive definite, then

$$\lambda_{\min}(\boldsymbol{P})\|\boldsymbol{x}\|^2 \leq \boldsymbol{x}^\top \boldsymbol{P}\boldsymbol{x} \leq \lambda_{\max}(\boldsymbol{P})\|\boldsymbol{x}\|^2,$$

where $\lambda_{\min}(\boldsymbol{P})$ denotes the smallest eigenvalue of $\boldsymbol{P}$, and $\lambda_{\max}(\boldsymbol{P})$ denotes the largest eigenvalue of $\boldsymbol{P}$.

Concepts from Geometry

4.1 Line Segments

4.2 Hyperplanes and Linear Varieties

4.3 Convex Sets

4.4 Neighborhoods

4.5 Polytopes and Polyhedra

第四章内容总结

超平面是 $n-1$ 维的, 因为 $x_1, \cdots, x_n$ 是 n 维的, 超平面给了一个限制条件, 所以有 $n-1$ 个自由度.

超平面法线的理解: 法线先记作 n (normal 的意思), normal 应满足同超平面内任意向量垂直, 那如何表示超平面内任何向量呢? 超平面内找一个已知点 a, 再找任意点 x, 则 $x-a$ 可代表超平面内任意向量, slide 中已证 $\langle u, a-x\rangle = 0$, 所以 u 即所求的法线.

超平面的交的理解方法, 方程每一个都可写成 $a_1 1 x_1 + ... + a_n x_n = b_1$, 每一个都是超平面, 所以解是超平面的交. 那么有几个超平面呢? 并不是有几个方程就是几个超平面, 因为有重复的

凸集概念在后面带限制条件的优化问题要用到, 直观上说, 优化迭代时, x_1 跳到 x_2, x_2 跳到 x_3, 我至少要保证点都在规定的集合中. 跳的时候还有跳多远 (步长) 的事, 如果两点间所有的点都在集合中, 可以根据需要, 有时跳近点.

今天：Elements of Calculus

5.1 Sequences and Limits

5.2 Differentiability

5.3 The Derivative Matrix

5.4 Differentiation Rules

5.5 Level Sets and Gradients

5.6 Taylor Series

5.1 Sequences and Limits ($\mathbb{R}^1$)

A *sequence of real numbers* is a function whose domain is the set of natural numbers $1, 2, \ldots, k, \ldots$ and whose range is contained in $\mathbb{R}$. Thus, a sequence of real numbers can be viewed as a set of numbers $\{x_1, x_2, \ldots, x_k, \ldots\}$, which is often also denoted as $\{x_k\}$ (or sometimes as $\{x_k\}_{k=1}^{\infty}$, to indicate explicitly the range of values that k can take).

A sequence $\{x_k\}$ is *increasing* if $x_1 < x_2 < \cdots < x_k \cdots$; that is, $x_k < x_{k+1}$ for all k. If $x_k \leq x_{k+1}$, then we say that the sequence is *nondecreasing*. Similarly, we can define *decreasing* and *nonincreasing sequences*. Nonincreasing or nondecreasing sequences are called *monotone sequences*.

A number $x^* \in \mathbb{R}$ is called the *limit* of the sequence $\{x_k\}$ if for any positive ε there is a number K (which may depend on ε) such that for all $k > K$, $|x_k - x^*| < \varepsilon$; that is, x_k lies between $x^* - \varepsilon$ and $x^* + \varepsilon$ for all $k > K$. In this case we write

$$x^* = \lim_{k \to \infty} x_k$$

or

$$x_k \to x^*.$$

5.1 Sequences and Limits ($\mathbb{R}^n$)

The notion of a sequence can be extended to sequences with elements in $\mathbb{R}^n$. Specifically, a sequence in $\mathbb{R}^n$ is a function whose domain is the set of natural numbers $1, 2, \ldots, k, \ldots$ and whose range is contained in $\mathbb{R}^n$. We use the notation $\{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots\}$ or $\{\boldsymbol{x}^{(k)}\}$ for sequences in $\mathbb{R}^n$. For limits of sequences in $\mathbb{R}^n$, we need to replace absolute values with vector norms. In other words, $\boldsymbol{x}^*$ is the limit of $\{\boldsymbol{x}^{(k)}\}$ if for any positive ε there is a number K (which may depend on ε) such that for all $k > K$, $\|\boldsymbol{x}^{(k)} - \boldsymbol{x}^*\| < \varepsilon$. As before, if a sequence $\{\boldsymbol{x}^{(k)}\}$ is convergent, we write $\boldsymbol{x}^* = \lim_{k\to\infty} \boldsymbol{x}^{(k)}$ or $\boldsymbol{x}^{(k)} \to \boldsymbol{x}^*$.

5.1 Sequences and Limits (唯一性)

Theorem 5.1 *A convergent sequence has only one limit.* □

Proof. We prove this result by contradiction. Suppose that a sequence $\{\boldsymbol{x}^{(k)}\}$ has two different limits, say $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. Then, we have $\|\boldsymbol{x}_1 - \boldsymbol{x}_2\| > 0$. Let

$$\varepsilon = \frac{1}{2}\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|.$$

From the definition of a limit, there exist K_1 and K_2 such that for $k > K_1$ we have $\|\boldsymbol{x}^{(k)} - \boldsymbol{x}_1\| < \varepsilon$, and for $k > K_2$ we have $\|\boldsymbol{x}^{(k)} - \boldsymbol{x}_2\| < \varepsilon$. Let $K = \max\{K_1, K_2\}$. Then, if $k > K$, we have $\|\boldsymbol{x}^{(k)} - \boldsymbol{x}_1\| < \varepsilon$ and $\|\boldsymbol{x}^{(k)} - \boldsymbol{x}_2\| < \varepsilon$. Adding $\|\boldsymbol{x}^{(k)} - \boldsymbol{x}_1\| < \varepsilon$ and $\|\boldsymbol{x}^{(k)} - \boldsymbol{x}_2\| < \varepsilon$ yields

$$\|\boldsymbol{x}^{(k)} - \boldsymbol{x}_1\| + \|\boldsymbol{x}^{(k)} - \boldsymbol{x}_2\| < 2\varepsilon.$$

Applying the triangle inequality gives

$$\begin{aligned} \|-\boldsymbol{x}_1 + \boldsymbol{x}_2\| &= \|\boldsymbol{x}^{(k)} - \boldsymbol{x}_1 - \boldsymbol{x}^{(k)} + \boldsymbol{x}_2\| \\ &= \|(\boldsymbol{x}^{(k)} - \boldsymbol{x}_1) - (\boldsymbol{x}^{(k)} - \boldsymbol{x}_2)\| \\ &\leq \|\boldsymbol{x}^{(k)} - \boldsymbol{x}_1\| + \|\boldsymbol{x}^{(k)} - \boldsymbol{x}_2\|. \end{aligned}$$

Therefore,

$$\|-\boldsymbol{x}_1 + \boldsymbol{x}_2\| = \|\boldsymbol{x}_1 - \boldsymbol{x}_2\| < 2\varepsilon.$$

However, this contradicts the assumption that $\|\boldsymbol{x}_1 - \boldsymbol{x}_2\| = 2\varepsilon$, which completes the proof. ∎

5.1 Sequences and Limits （有界性）

A sequence $\{\boldsymbol{x}^{(k)}\}$ in $\mathbb{R}^n$ is *bounded* if there exists a number $B \geq 0$ such that $\|\boldsymbol{x}^{(k)}\| \leq B$ for all $k = 1, 2, \ldots$.

Theorem 5.2 *Every convergent sequence is bounded.* □

Proof. Let $\{\boldsymbol{x}^{(k)}\}$ be a convergent sequence with limit $\boldsymbol{x}^*$. Choose $\varepsilon = 1$. Then, by definition of the limit, there exists a natural number K such that for all $k > K$,

$$\|\boldsymbol{x}^{(k)} - \boldsymbol{x}^*\| < 1.$$

By the result of Exercise 2.9, we get

$$\|\boldsymbol{x}^{(k)}\| - \|\boldsymbol{x}^*\| \leq \|\boldsymbol{x}^{(k)} - \boldsymbol{x}^*\| < 1 \quad \text{for all } k > K.$$

Therefore,

$$\|\boldsymbol{x}^{(k)}\| < \|\boldsymbol{x}^*\| + 1 \quad \text{for all } k > K.$$

Letting

$$B = \max\left\{\|\boldsymbol{x}^{(1)}\|, \|\boldsymbol{x}^{(2)}\|, \ldots, \|\boldsymbol{x}^{(K)}\|, \|\boldsymbol{x}^*\| + 1\right\},$$

we have

$$B \geq \|\boldsymbol{x}^{(k)}\| \quad \text{for all } k,$$

which means that the sequence $\{\boldsymbol{x}^{(k)}\}$ is bounded. ∎

5.1 Sequences and Limits （上确界）

For a sequence $\{x_k\}$ in $\mathbb{R}$, a number B is called an *upper bound* if $x_k \leq B$ for all $k = 1, 2, \ldots$. In this case, we say that $\{x_k\}$ is *bounded above*. Similarly, B is called a *lower bound* if $x_k \geq B$ for all $k = 1, 2, \ldots$. In this case, we say that $\{x_k\}$ is *bounded below*. Clearly, a sequence is bounded if it is both bounded above and bounded below.

Any sequence $\{x_k\}$ in $\mathbb{R}$ that has an upper bound has a *least upper bound* (also called the *supremum*), which is the smallest number B that is an upper bound of $\{x_k\}$. Similarly, any sequence $\{x_k\}$ in $\mathbb{R}$ that has a lower bound has a *greatest lower bound* (also called the *infimum*). If B is the least upper bound of the sequence $\{x_k\}$, then $x_k \leq B$ for all k, and for any $\varepsilon > 0$, there exists a number K such that $x_K > B - \varepsilon$. An analogous statement applies to the greatest lower bound: If B is the greatest lower bound of $\{x_k\}$, then $x_k \geq B$ for all k, and for any $\varepsilon > 0$, there exists a number K such that $x_K < B + \varepsilon$.

5.1 Sequences and Limits （单调有界必收敛）

Theorem 5.3 *Every monotone bounded sequence in $\mathbb{R}$ is convergent.* □

Proof. We prove the theorem for nondecreasing sequences. The proof for nonincreasing sequences is analogous.

Let $\{x_k\}$ be a bounded nondecreasing sequence in $\mathbb{R}$ and x^* the least upper bound. Fix a number $\varepsilon > 0$. Then, there exists a number K such that $x_K > x^* - \varepsilon$. Because $\{x_k\}$ is nondecreasing, for any $k \geq K$,

$$x_k \geq x_K > x^* - \varepsilon.$$

Also, because x^* is an upper bound of $\{x_k\}$, we have

$$x_k \leq x^* < x^* + \varepsilon.$$

Therefore, for any $k \geq K$,

$$|x_k - x^*| < \varepsilon,$$

which means that $x_k \to x^*$. ∎

5.1 Sequences and Limits （收敛序列的子序列必收敛）

Theorem 5.4 *Consider a convergent sequence* $\{\boldsymbol{x}^{(k)}\}$ *with limit* $\boldsymbol{x}^*$. *Then, any subsequence of* $\{\boldsymbol{x}^{(k)}\}$ *also converges to* $\boldsymbol{x}^*$. □

Proof. Let $\{\boldsymbol{x}^{(m_k)}\}$ be a subsequence of $\{\boldsymbol{x}^{(k)}\}$, where $\{m_k\}$ is an increasing sequence of natural numbers. Observe that $m_k \geq k$ for all $k = 1, 2, \ldots$. To show this, first note that $m_1 \geq 1$ because m_1 is a natural number. Next, we proceed by induction by assuming that $m_k \geq k$. Then, we have $m_{k+1} > m_k \geq k$, which implies that $m_{k+1} \geq k + 1$. Therefore, we have shown that $m_k \geq k$ for all $k = 1, 2, \ldots$.

Let $\varepsilon > 0$ be given. Then, by definition of the limit, there exists K such that $\|\boldsymbol{x}^{(k)} - \boldsymbol{x}^*\| < \varepsilon$ for any $k > K$. Because $m_k \geq k$, we also have $\|\boldsymbol{x}^{(m_k)} - \boldsymbol{x}^*\| < \varepsilon$ for any $k > K$. This means that

$$\lim_{k \to \infty} \boldsymbol{x}^{(m_k)} = \boldsymbol{x}^*.$$

∎

It turns out that any bounded sequence contains a convergent subsequence. This result is called the *Bolzano-Weierstrass theorem* (see [2, p. 70]).

5.1 Sequences and Limits （连续）

Consider a function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ and a point $\boldsymbol{x}_0 \in \mathbb{R}^n$. Suppose that there exists $\boldsymbol{f}^*$ such that for any convergent sequence $\{\boldsymbol{x}^{(k)}\}$ with limit $\boldsymbol{x}_0$, we have

$$\lim_{k\to\infty} \boldsymbol{f}(\boldsymbol{x}^{(k)}) = \boldsymbol{f}^*.$$

Then, we use the notation

$$\lim_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{f}(\boldsymbol{x})$$

to represent the limit $\boldsymbol{f}^*$.

It turns out that $\boldsymbol{f}$ is continuous at $\boldsymbol{x}_0$ if and only if for any convergent sequence $\{\boldsymbol{x}^{(k)}\}$ with limit $\boldsymbol{x}_0$, we have

$$\lim_{k\to\infty} \boldsymbol{f}(\boldsymbol{x}^{(k)}) = \boldsymbol{f}\left(\lim_{k\to\infty} \boldsymbol{x}^{(k)}\right) = \boldsymbol{f}(\boldsymbol{x}_0)$$

(see [2, p. 137]). Therefore, using the notation introduced above, the function $\boldsymbol{f}$ is continuous at $\boldsymbol{x}_0$ if and only if

$$\lim_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x}_0).$$

5.1 Sequences and Limits

矩阵的极限

We say that a sequence $\{\boldsymbol{A}_k\}$ of $m \times n$ matrices converges to the $m \times n$ matrix $\boldsymbol{A}$ if

$$\lim_{k \to \infty} \|\boldsymbol{A} - \boldsymbol{A}_k\| = 0.$$

Lemma 5.1 *Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$. Then, $\lim_{k \to \infty} \boldsymbol{A}^k = \boldsymbol{O}$ if and only if the eigenvalues of $\boldsymbol{A}$ satisfy $|\lambda_i(\boldsymbol{A})| < 1$, $i = 1, \ldots, n$.* □

Proof. To prove this theorem, we use the *Jordan form* (see, e.g., [47]). Specifically, it is well known that any square matrix is similar to the Jordan form: There exists a nonsingular $\boldsymbol{T}$ such that

$$\boldsymbol{T}\boldsymbol{A}\boldsymbol{T}^{-1} = \operatorname{diag}[\boldsymbol{J}_{m_1}(\lambda_1), \ldots, \boldsymbol{J}_{m_s}(\lambda_1), \boldsymbol{J}_{n_1}(\lambda_2), \ldots, \boldsymbol{J}_{t_\nu}(\lambda_q)] \triangleq \boldsymbol{J},$$

5.2 Differentiability

Differential calculus is based on the idea of approximating an arbitrary function by an *affine function*. A function $\mathcal{A} : \mathbb{R}^n \to \mathbb{R}^m$ is *affine* if there exists a *linear function* $\mathcal{L} : \mathbb{R}^n \to \mathbb{R}^m$ and a vector $\boldsymbol{y} \in \mathbb{R}^m$ such that

$$\mathcal{A}(\boldsymbol{x}) = \mathcal{L}(\boldsymbol{x}) + \boldsymbol{y}$$

for every $\boldsymbol{x} \in \mathbb{R}^n$. Consider a function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ and a point $\boldsymbol{x}_0 \in \mathbb{R}^n$. We wish to find an affine function $\mathcal{A}$ that approximates $\boldsymbol{f}$ near the point $\boldsymbol{x}_0$. First, it is natural to impose the condition

$$\mathcal{A}(\boldsymbol{x}_0) = \boldsymbol{f}(\boldsymbol{x}_0).$$

Because $\mathcal{A}(\boldsymbol{x}) = \mathcal{L}(\boldsymbol{x}) + \boldsymbol{y}$, we obtain $\boldsymbol{y} = \boldsymbol{f}(\boldsymbol{x}_0) - \mathcal{L}(\boldsymbol{x}_0)$. By the linearity of $\mathcal{L}$,

$$\mathcal{L}(\boldsymbol{x}) + \boldsymbol{y} = \mathcal{L}(\boldsymbol{x}) - \mathcal{L}(\boldsymbol{x}_0) + \boldsymbol{f}(\boldsymbol{x}_0) = \mathcal{L}(\boldsymbol{x} - \boldsymbol{x}_0) + \boldsymbol{f}(\boldsymbol{x}_0).$$

Hence, we may write

$$\mathcal{A}(\boldsymbol{x}) = \mathcal{L}(\boldsymbol{x} - \boldsymbol{x}_0) + \boldsymbol{f}(\boldsymbol{x}_0).$$

Next, we require that $\mathcal{A}(\boldsymbol{x})$ approaches $\boldsymbol{f}(\boldsymbol{x})$ faster than $\boldsymbol{x}$ approaches $\boldsymbol{x}_0$; that is,

$$\lim_{\boldsymbol{x} \to \boldsymbol{x}_0, \boldsymbol{x} \in \Omega} \frac{\|\boldsymbol{f}(\boldsymbol{x}) - \mathcal{A}(\boldsymbol{x})\|}{\|\boldsymbol{x} - \boldsymbol{x}_0\|} = 0.$$

The conditions above on $\mathcal{A}$ ensure that $\mathcal{A}$ approximates $\boldsymbol{f}$ near $\boldsymbol{x}_0$ in the sense that the error in the approximation at a given point is "small" compared with the distance of the point from $\boldsymbol{x}_0$.

5.3 The Derivative Matrix (1/4)

Any linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$, and in particular the derivative $\mathcal{L}$ of $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, can be represented by an $m \times n$ matrix. To find the matrix representation $\boldsymbol{L}$ of the derivative $\mathcal{L}$ of a differentiable function $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, we use the natural basis $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$ for $\mathbb{R}^n$. Consider the vectors

$$\boldsymbol{x}_j = \boldsymbol{x}_0 + t\boldsymbol{e}_j, \quad j = 1, \ldots, n.$$

By the definition of the derivative, we have

$$\lim_{t \to 0} \frac{\boldsymbol{f}(\boldsymbol{x}_j) - (t\boldsymbol{L}\boldsymbol{e}_j + \boldsymbol{f}(\boldsymbol{x}_0))}{t} = \boldsymbol{0}$$

for $j = 1, \ldots, n$. This means that

$$\lim_{t \to 0} \frac{\boldsymbol{f}(\boldsymbol{x}_j) - \boldsymbol{f}(\boldsymbol{x}_0)}{t} = \boldsymbol{L}\boldsymbol{e}_j$$

for $j = 1, \ldots, n$. But $\boldsymbol{L}\boldsymbol{e}_j$ is the jth column of the matrix $\boldsymbol{L}$. On the other hand, the vector $\boldsymbol{x}_j$ differs from $\boldsymbol{x}_0$ only in the jth coordinate, and in that coordinate the difference is just the number t. Therefore, the left side of the preceding equation is the partial derivative

$$\frac{\partial \boldsymbol{f}}{\partial x_j}(\boldsymbol{x}_0).$$

Because vector limits are computed by taking the limit of each coordinate function, it follows that if

$$\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} f_1(\boldsymbol{x}) \\ \vdots \\ f_m(\boldsymbol{x}) \end{bmatrix},$$

then

$$\frac{\partial \boldsymbol{f}}{\partial x_j}(\boldsymbol{x}_0) = \begin{bmatrix} \frac{\partial f_1}{\partial x_j}(\boldsymbol{x}_0) \\ \vdots \\ \frac{\partial f_m}{\partial x_j}(\boldsymbol{x}_0) \end{bmatrix},$$

5.3 The Derivative Matrix (2/4)

Theorem 5.5 *If a function* $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ *is differentiable at* $\boldsymbol{x}_0$*, then the derivative of* $\boldsymbol{f}$ *at* $\boldsymbol{x}_0$ *is determined uniquely and is represented by the* $m \times n$ *derivative matrix* $D\boldsymbol{f}(\boldsymbol{x}_0)$*. The best affine approximation to* $\boldsymbol{f}$ *near* $\boldsymbol{x}_0$ *is then given by*

$$\mathcal{A}(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x}_0) + D\boldsymbol{f}(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0),$$

in the sense that

$$\boldsymbol{f}(\boldsymbol{x}) = \mathcal{A}(\boldsymbol{x}) + \boldsymbol{r}(\boldsymbol{x})$$

and $\lim_{\boldsymbol{x} \to \boldsymbol{x}_0} \|\boldsymbol{r}(\boldsymbol{x})\| / \|\boldsymbol{x} - \boldsymbol{x}_0\| = 0$*. The columns of the derivative matrix* $D\boldsymbol{f}(\boldsymbol{x}_0)$ *are vector partial derivatives. The vector*

$$\frac{\partial \boldsymbol{f}}{\partial x_j}(\boldsymbol{x}_0)$$

is a tangent vector at $\boldsymbol{x}_0$ *to the curve* $\boldsymbol{f}$ *obtained by varying only the* j*th coordinate of* $\boldsymbol{x}$*.* □

5.3 The Derivative Matrix

If $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable, then the function ∇f defined by

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\boldsymbol{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\boldsymbol{x}) \end{bmatrix} = Df(\boldsymbol{x})^\top$$

is called the *gradient* of f. The gradient is a function from $\mathbb{R}^n$ to $\mathbb{R}^n$, and can be pictured as a *vector field*, by drawing the arrow representing $\nabla f(\boldsymbol{x})$ so that its tail starts at $\boldsymbol{x}$.

Given $f : \mathbb{R}^n \to \mathbb{R}$, if ∇f is differentiable, we say that f is *twice differentiable*, and we write the derivative of ∇f as

$$D^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \frac{\partial^2 f}{\partial x_2 \partial x_n} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

(The notation $\frac{\partial^2 f}{\partial x_i \partial x_j}$ represents taking the partial derivative of f with respect to x_j first, then with respect to x_i.) The matrix $D^2 f(\boldsymbol{x})$ is called the *Hessian matrix* of f at $\boldsymbol{x}$, and is often also denoted $\boldsymbol{F}(\boldsymbol{x})$.

Note that the Hessian matrix of a function $f : \mathbb{R}^n \to \mathbb{R}$ at $\boldsymbol{x}$ is symmetric if f is twice continuously differentiable at $\boldsymbol{x}$. This is a well-known result from calculus called *Clairaut's theorem* or *Schwarz's theorem.* However, if the second partial derivatives of f are not continuous, then there is no guarantee that the Hessian is symmetric, as shown in the following well-known example.

Example 5.1 Consider the function

$$f(\boldsymbol{x}) = \begin{cases} x_1 x_2 (x_1^2 - x_2^2)/(x_1^2 + x_2^2) & \text{if } \boldsymbol{x} \neq \mathbf{0} \\ 0 & \text{if } \boldsymbol{x} = \mathbf{0}. \end{cases}$$

5.4 Differentiation Rules

We now introduce the *chain rule* for differentiating the composition $g(\boldsymbol{f}(t))$, of a function $\boldsymbol{f} : \mathbb{R} \to \mathbb{R}^n$ and a function $g : \mathbb{R}^n \to \mathbb{R}$.

Theorem 5.6 *Let $g : \mathcal{D} \to \mathbb{R}$ be differentiable on an open set $\mathcal{D} \subset \mathbb{R}^n$, and let $\boldsymbol{f} : (a,b) \to \mathcal{D}$ be differentiable on (a,b). Then, the composite function $h : (a,b) \to \mathbb{R}$ given by $h(t) = g(\boldsymbol{f}(t))$ is differentiable on (a,b), and*

$$h'(t) = Dg(\boldsymbol{f}(t))D\boldsymbol{f}(t) = \nabla g(\boldsymbol{f}(t))^\top \begin{bmatrix} f_1'(t) \\ \vdots \\ f_n'(t) \end{bmatrix}.$$

□

Proof. By definition,

$$h'(t) = \lim_{s \to t} \frac{h(s) - h(t)}{s - t} = \lim_{s \to t} \frac{g(\boldsymbol{f}(s)) - g(\boldsymbol{f}(t))}{s - t}$$

if the limit exists. By Theorem 5.5 we write

$$g(\boldsymbol{f}(s)) - g(\boldsymbol{f}(t)) = Dg(\boldsymbol{f}(t))(\boldsymbol{f}(s) - \boldsymbol{f}(t)) + r(s),$$

where $\lim_{s \to t} r(s)/(s-t) = 0$. Therefore,

$$\frac{h(s) - h(t)}{s - t} = Dg(\boldsymbol{f}(t))\frac{\boldsymbol{f}(s) - \boldsymbol{f}(t)}{s - t} + \frac{r(s)}{s - t}.$$

Letting $s \to t$ yields

$$h'(t) = \lim_{s \to t} Dg(\boldsymbol{f}(t))\frac{\boldsymbol{f}(s) - \boldsymbol{f}(t)}{s - t} + \frac{r(s)}{s - t} = Dg(\boldsymbol{f}(t))D\boldsymbol{f}(t).$$

∎

5.4 Differentiation Rules

Next, we present the *product rule*. Let $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ and $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}^m$ be two differentiable functions. Define the function $h : \mathbb{R}^n \to \mathbb{R}$ by $h(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x})^\top \boldsymbol{g}(\boldsymbol{x})$. Then, h is also differentiable and

$$Dh(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x})^\top D\boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})^\top D\boldsymbol{f}(\boldsymbol{x}).$$

We end this section with a list of some useful formulas from multivariable calculus. In each case, we compute the derivative with respect to $\boldsymbol{x}$. Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a given matrix and $\boldsymbol{y} \in \mathbb{R}^m$ a given vector. Then,

$$\begin{aligned} D(\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{x}) &= \boldsymbol{y}^\top \boldsymbol{A} \\ D(\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}) &= \boldsymbol{x}^\top (\boldsymbol{A} + \boldsymbol{A}^\top) \quad \text{if } m = n. \end{aligned}$$

It follows from the first formula above that if $\boldsymbol{y} \in \mathbb{R}^n$, then

$$D(\boldsymbol{y}^\top \boldsymbol{x}) = \boldsymbol{y}^\top.$$

It follows from the second formula above that if $\boldsymbol{Q}$ is a symmetric matrix, then

$$D(\boldsymbol{x}^\top \boldsymbol{Q}\boldsymbol{x}) = 2\boldsymbol{x}^\top \boldsymbol{Q}.$$

In particular,

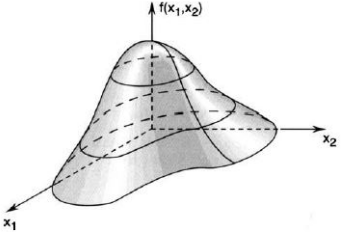
$$D(\boldsymbol{x}^\top \boldsymbol{x}) = 2\boldsymbol{x}^\top.$$

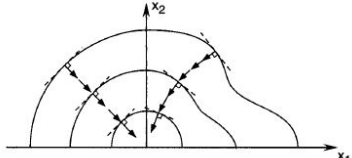
5.5 Level Sets and Gradients (1/3)

The *level set* of a function $f : \mathbb{R}^n \to \mathbb{R}$ at level c is the set of points

$$S = \{\boldsymbol{x} : f(\boldsymbol{x}) = c\}.$$

For $f : \mathbb{R}^2 \to \mathbb{R}$, we are usually interested in S when it is a curve. For $f : \mathbb{R}^3 \to \mathbb{R}$, the sets S most often considered are surfaces.



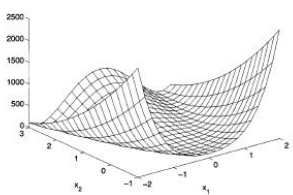


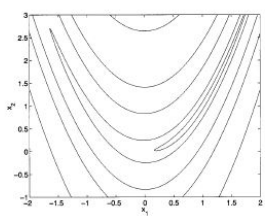
5.5 Level Sets and Gradients (2/3)

Example 5.2 Consider the following real-valued function on $\mathbb{R}^2$:

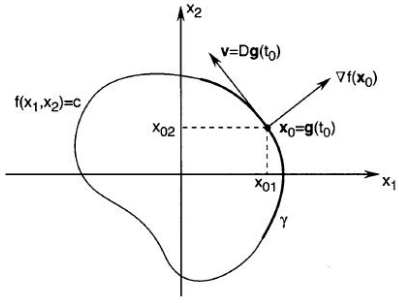
$$f(\boldsymbol{x}) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2, \qquad \boldsymbol{x} = [x_1, x_2]^\top.$$

The function above is called *Rosenbrock's function*. A plot of the function f is shown in Figure 5.2. The level sets of f at levels 0.7, 7, 70, 200, and 700 are depicted in Figure 5.3. These level sets have a particular shape resembling





5.5 Level Sets and Gradients (3/3)



To say that a point $\boldsymbol{x}_0$ is on the level set S at level c means that $f(\boldsymbol{x}_0) = c$. Now suppose that there is a curve γ lying in S and parameterized by a continuously differentiable function $\boldsymbol{g} : \mathbb{R} \to \mathbb{R}^n$. Suppose also that $\boldsymbol{g}(t_0) = \boldsymbol{x}_0$ and $D\boldsymbol{g}(t_0) = \boldsymbol{v} \neq \boldsymbol{0}$, so that $\boldsymbol{v}$ is a tangent vector to γ at $\boldsymbol{x}_0$ (see Figure 5.4). Applying the chain rule to the function $h(t) = f(\boldsymbol{g}(t))$ at t_0 gives

$$h'(t_0) = Df(\boldsymbol{g}(t_0))D\boldsymbol{g}(t_0) = Df(\boldsymbol{x}_0)\boldsymbol{v}.$$

But since γ lies on S, we have

5.6 Taylor Series

演示: https://www.geogebra.org/m/rfdkjwpx

Theorem 5.8 ***Taylor's Theorem.*** *Assume that a function $f : \mathbb{R} \to \mathbb{R}$ is m times continuously differentiable (i.e., $f \in C^m$) on an interval $[a, b]$. Denote $h = b - a$. Then,*

$$f(b) = f(a) + \frac{h}{1!}f^{(1)}(a) + \frac{h^2}{2!}f^{(2)}(a) + \cdots + \frac{h^{m-1}}{(m-1)!}f^{(m-1)}(a) + R_m,$$

(called Taylor's formula) where $f^{(i)}$ is the ith derivative of f, and

$$R_m = \frac{h^m(1-\theta)^{m-1}}{(m-1)!}f^{(m)}(a+\theta h) = \frac{h^m}{m!}f^{(m)}(a+\theta' h),$$

with $\theta, \theta' \in (0, 1)$.

5.6 Taylor Series

Theorem 5.9 *If a function* $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ *is differentiable on an open set* $\Omega \subset \mathbb{R}^n$, *then for any pair of points* $\boldsymbol{x}, \boldsymbol{y} \in \Omega$, *there exists a matrix* $\boldsymbol{M}$ *such that*

$$\boldsymbol{f}(\boldsymbol{x}) - \boldsymbol{f}(\boldsymbol{y}) = \boldsymbol{M}(\boldsymbol{x} - \boldsymbol{y}).$$

□

The mean value theorem follows from Taylor's theorem (for the case where $m = 1$) applied to each component of $\boldsymbol{f}$. It is easy to see that $\boldsymbol{M}$ is a matrix whose rows are the rows of $D\boldsymbol{f}$ evaluated at points that lie on the line segment joining $\boldsymbol{x}$ and $\boldsymbol{y}$ (these points may differ from row to row).

优化： Topic 1: Introduction

★ Problem

$$\min_{\boldsymbol{x}\in\Omega} f(\boldsymbol{x})$$

- f: objective function
- $\Omega \subset \mathbb{R}^n$: feasible set / constraint set

$\max f(\boldsymbol{x})$ can be changed to $\min -f(\boldsymbol{x})$

★ Optimization

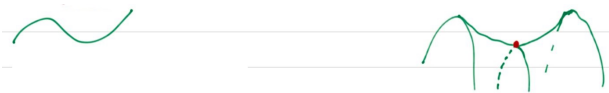
1. **Modeling:** Practical problems $\rightarrow$ Optimization problems
2. **Algorithms:** Methods to solve
3. **Software:** Implement

Topic 1: Introduction

★ Types of Optimization Problems

- Constrained vs. Non-Constrained
  $\boldsymbol{x} \in \Omega$ $\boldsymbol{x} \in \mathbb{R}^n$, i.e., $\boldsymbol{x}$ is free
- Convex vs. Non-convex
  both f and Ω are convex
- Smooth vs. Non-Smooth
  ∇f ∂f
- 1-d $\mathbb{R}^1$ vs. n-d $\mathbb{R}^n$

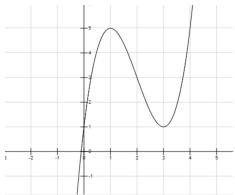
{ illustration, motivation, understanding } Harder due to infinite directions



Topic 1: Introduction

★ Examples

1. $\min_{x\in[0,5]} f(x), \quad f(x) = x^3 - 6x^2 + 9x + 1$



This is a 1-d smooth, non-convex, and constrained optimization

sol. (only outline, not detailed solution)

critical pts: $f'(x) = 0, \quad x = 1 \quad$ or $\quad x = 3$

min or max ? $f''(1) < 0 \quad$ local max, $\qquad f''(3) > 0 \quad$ local min

★ Examples

Question: how about the case $f''(x) = 0$?

Can we conclude the answer is $x = 3$?

No! This is a constrained opt. problem. Check

1. $x = 3 \in \Omega = [0, 5]$,
2. compare with values on boundary: $f(0), f(5)$

The final answer is $\min_{x \in [0,5]} f(x) = 1$, and is achieved at $x = 0$, 3.

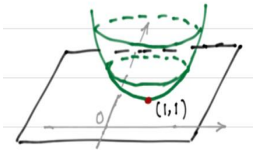
Question: Change the constraint $\min_x f(x)$ s.t. $x \in [-1, 1]$

★ **Examples**

2.

$$f(x,y) = (x-1)^2 + (y-1)^2$$

$$\min_{(x,y)\in\mathbb{R}^2} f(x,y)$$



Critical pts:

$$\begin{cases} \frac{\partial f}{\partial x}(x,y) = 0, & 2(x-1) = 0 \quad \Rightarrow \quad x = 1 \\ \frac{\partial f}{\partial y}(x,y) = 0, & 2(y-1) = 0 \quad \Rightarrow \quad y = 1 \end{cases}$$

Topic 1: Introduction

★ **Examples**

Second order conditions.

- 1-d: $f''(x) > 0$
- 2-d: Hessian matrix

$$\nabla^2 f = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}, \text{ formally } \nabla^2 f \succ 0$$

For a symmetric matrix $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$,

$$A \succ 0 \text{ if } \quad (u, v) A \begin{pmatrix} u \\ v \end{pmatrix} > 0 \quad \forall (u, v) \in \mathbb{R}^2,$$

i.e., $\alpha_{11} u^2 + 2 a_{12} uv + a_{22} v^2 > 0, \quad \forall u, v$

特殊值: Set $v = 0, u = 1 \quad \rightarrow \quad a_{11} > 0$

Set $u = 0, v = 1 \quad \rightarrow \quad a_{22} > 0$

$\det(A) = a_{11} a_{22} - a_{12}^2 > 0$

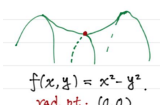
Topic 1: Introduction

2×2 symmetric $A \succ 0$ if a_{11} and $\det(A) > 0$

A can be better viewed as $A = Q^T \bigwedge Q$, where $\bigwedge = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ and Q consists of corresponding eigenvectors, and $Q^T Q = I$

$A \succ 0 \Leftrightarrow \lambda_1 > 0, \lambda_2 > 0$

$$A \not\succ 0 \begin{cases} \lambda_1 \leq 0, \lambda_2 \leq 0 \\ \lambda_1 \lambda_2 \leq 0 \quad (+,-) \quad \text{saddle pt} \end{cases}$$

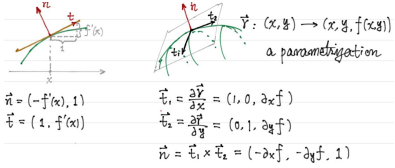


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Notation. $f : \mathbb{R}^n \to \mathbb{R} \quad \boldsymbol{x} \in \mathbb{R}^n, \quad \boldsymbol{x} = (x_1, x_2, ..., x_n)$

$$Df \triangleq (\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, ..., \frac{\partial f}{\partial x_n}), \quad \nabla f = (Df)^T = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ ... \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

特例: 1-d, $f'(x)$ is the slope of the tangent line $\frac{\partial f}{\partial x_n}$ at x



$H = \nabla^2 f = (\frac{\partial^2 f}{\partial x_i \partial x_j})$. As $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$, ∇^2 is symmetric

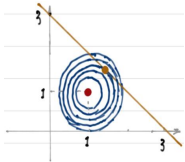
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★ Examples

3. $\min_{x,y} f(x,y)$ s.t. $x+y=3$

$f(x,y)=(x-1)^2+(y-1)^2, \quad \Omega=\{(x,y)\in\mathbb{R}^2 | x+y=3\}$

This is a 2-d smooth, convex and constrained optimization.



Level set $S_c=\{\boldsymbol{x}: \quad f(\boldsymbol{x})=c\}$

Level set S_c is a curve in $\mathbb{R}^2$

Graph of $f \quad (\boldsymbol{x}, f(\boldsymbol{x}))\in\mathbb{R}^3$ is a surface in $\mathbb{R}^3$

Non-constrained minimum is at $(1,1)$ red pt $\nabla f(1,1)=0$

With an equality constraint, the minimum is changed to brown where $\nabla f \neq 0$

★ **Examples**

For this problem, we can eliminate y to get a 1-d non-constraint smooth and convex opt problem.

$$y = 3 - x \quad \tilde{f}(x) \triangleq f(x, 3-x) = (x-1)^2 + (x-2)^2$$

$$\tilde{f}'(x) = 0, \text{ so } x = \frac{3}{2}, \quad y = \frac{3}{2} \quad \tilde{f}''(x) = 4 > 0 \forall x$$

$x = \frac{3}{2}$ is a local minimum, and as $\tilde{f}$ is convex, it is a global min. So $\min f(x, y)$ s.t. $x + y = 3$ is $\frac{1}{2}$, and the minimum pt $(\frac{3}{2}, \frac{3}{2})$

Fact: for a convex function, a local minimum is also a global one (to be proved soon)

Topic 1: Introduction

★ level sets and gradient

$S_c = \{\boldsymbol{x} | f(\boldsymbol{x}) = c\}$. This is a smooth curve for most c.

How to represent/describe a curve $f(\boldsymbol{x}) = c$? Parametrization.

$g: \quad \mathbb{R} \quad \to \quad \mathbb{R}^n \quad g(t) = (x(t), y(t))$ in $\mathbb{R}^2$

$h: \quad \mathbb{R} \quad \to \quad \mathbb{R} \quad h(t) \triangleq f(g(t))$

$h(t) = c$ by definition. So $h'(t) = 0$

By chain rule, $h'(t) = \nabla f(g(t)) \cdot g'(t)$. So for a pt $\boldsymbol{x_0} \in S_c$, we have $\nabla f(\boldsymbol{x_0}) \cdot \boldsymbol{v} = 0$ where $\boldsymbol{v}$ is a tangent vector of S_c at $\boldsymbol{x}_0$.

Theorem. $\nabla f(\boldsymbol{x_0}) \perp \boldsymbol{v}, \quad \forall \boldsymbol{v}$ tangent vector at $\boldsymbol{x}_0$ of the level set S_c for $c = f(\boldsymbol{x_0})$

$\nabla f(\boldsymbol{x})$ is the direction of maximum rate of increase of f at $\boldsymbol{x}$

$-\nabla f(\boldsymbol{x})$ is the direction of maximum rate of decrease of f at $\boldsymbol{x}$

$-\nabla f(x)$: steepest descent direction

Topic 1: Introduction

★ level sets and gradient

$S_c = \{\boldsymbol{x} | f(\boldsymbol{x}) = c\}$. This is a smooth curve for most c.

How to represent/describe a curve $f(\boldsymbol{x}) = c$? Parametrization.

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$-\nabla f(\boldsymbol{x})$ is the direction of maximum rate of decrease of f at $\boldsymbol{x}$

$-\nabla f(x)$: steepest descent direction

Topic 1: Introduction

★ **Examples**

4. Rosenbrock function $f(x_1, x_2) = 100(x_2 - {x_1}^2)^2 + (1 - x_1)^2$

$$\min_{\boldsymbol{x} \in \mathbb{R}^2} f$$

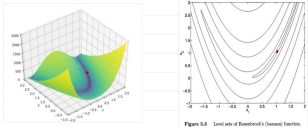
non-constrained, smooth, but non-convex

$\nabla f(\boldsymbol{x}) = \begin{pmatrix} -400x_1(x_2 - {x_1}^2) - 2(1 - x_1) \\ 200(x_2 - {x_1}^2) \end{pmatrix}$, critical point $(1, 1)$

$H(\boldsymbol{x}) = \nabla^2 f(\boldsymbol{x}) = \begin{pmatrix} 1200{x_1}^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}$

$H(1, 1) = \begin{pmatrix} 802 & -400 \\ -400 & 200 \end{pmatrix}$, so $(1, 1)$ is a local minimum.

$(1, 1)$ is also a global minimum. It is inside a long, narrow, parabolic shaped flab valley.



作业：习题

6.1 Consider the problem

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in \Omega, \end{aligned}$$

where $f \in \mathcal{C}^2$. For each of the following specifications for Ω, $\boldsymbol{x}^*$, and f, determine if the given point $\boldsymbol{x}^*$ is: (i) definitely a local minimizer; (ii) definitely not a local minimizer; or (iii) possibly a local minimizer.

b. $f : \mathbb{R}^2 \to \mathbb{R}$, $\Omega = \{\boldsymbol{x} = [x_1, x_2]^\top : x_1 \geq 1, x_2 \geq 2\}$, $\boldsymbol{x}^* = [1, 2]^\top$, and gradient $\nabla f(\boldsymbol{x}^*) = [1, 0]^\top$.

习题

6.1 Consider the problem

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in \Omega, \end{aligned}$$

where $f \in \mathcal{C}^2$. For each of the following specifications for Ω, $\boldsymbol{x}^\star$, and f, determine if the given point $\boldsymbol{x}^\star$ is: (i) definitely a local minimizer; (ii) definitely not a local minimizer; or (iii) possibly a local minimizer.

c. $f : \mathbb{R}^2 \to \mathbb{R}$, $\Omega = \{\boldsymbol{x} = [x_1, x_2]^\top : x_1 \geq 0, x_2 \geq 0\}$, $\boldsymbol{x}^\star = [1, 2]^\top$, gradient $\nabla f(\boldsymbol{x}^\star) = [0, 0]^\top$, and Hessian $\boldsymbol{F}(\boldsymbol{x}^\star) = \boldsymbol{I}$ (identity matrix).

d. $f : \mathbb{R}^2 \to \mathbb{R}$, $\Omega = \{\boldsymbol{x} = [x_1, x_2]^\top : x_1 \geq 1, x_2 \geq 2\}$, $\boldsymbol{x}^\star = [1, 2]^\top$, gradient $\nabla f(\boldsymbol{x}^\star) = [1, 0]^\top$, and Hessian

$$\boldsymbol{F}(\boldsymbol{x}^\star) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

知识点、重点、难点

知识点

- 优化问题的一般形式：

$$\min_{\boldsymbol{x}\in\Omega} f(\boldsymbol{x})$$

  其中 $f(\boldsymbol{x})$ 为目标函数，Ω 为约束集合。
- 优化问题的主要分类：有约束 / 无约束、凸 / 非凸、光滑 / 非光滑。
- 局部最小点与全局最小点的概念。
- 梯度（Gradient）与 Hessian 矩阵在多维函数中的几何意义。

知识点、重点、难点

重点

优化问题的核心在于通过一阶条件 $\nabla f(x) = 0$ 寻找候选极值点，并利用二阶条件（Hessian 的正定性）判断其为极小点；凸函数中局部最小即为全局最小，而含约束问题则常可通过变量消元化为无约束形式。

难点

- 区分局部最小点、鞍点与约束最小点。
- 约束优化中如何判断最优解是否在边界上。

一句话总结：

本节核心在于理解优化问题的建模、分类与基本求解原则，掌握梯度和 Hessian 在判断极值中的作用，为后续最优条件的学习奠定基础。